1) Smart Online Examination Registration System:-

a) Input conditions and Equivalence clauses:-

Input	valid class	Invalid class
course reg	Registered	NOT Registered
Attendence	75% - 100%	Below 75% above 100%
Exam fee	paid	Unpaid
Register no	Exactly 12 digit	Less or more than 12 digits, non-numeric

Expected output:

* All conditions valid: Registration Accepted
* Attendence below 75%: Registration Rejected
* Fee unpaid: Rayment Requested.
* Invalid register number: Error Msg.

b) ECP and BVA.

Test case	Attendence	Expected output
TC1	74%	Rejected
TC2	75%	Accepted if other condition are valid
TC3	76%	Accepted if other condition are valid.

c) Decision Table:-

course Registered	Attendence ≥75%	Fee Paid
yes	yes	yes
NO	yes	yes
yes	NO	yes
yes	yes	NO
yes	yes	yes

d) Random Testing:-

Test case	Attendence	Reg. NO	Expected Output.
TC1	50 %.	123456789012	Rejected
TC2	80%.	123456789012	Accepted if other condition are valid
TC 3	100%.	12345678901 2	Invalid attendance
TC4	90%.	12345	Error message.

Random testing uses different inputs to discover unexpected defects that fixed test case may miss.

e) Risk - Based Testing:-

1) Test important conditions such as attendence, payment, register number first.

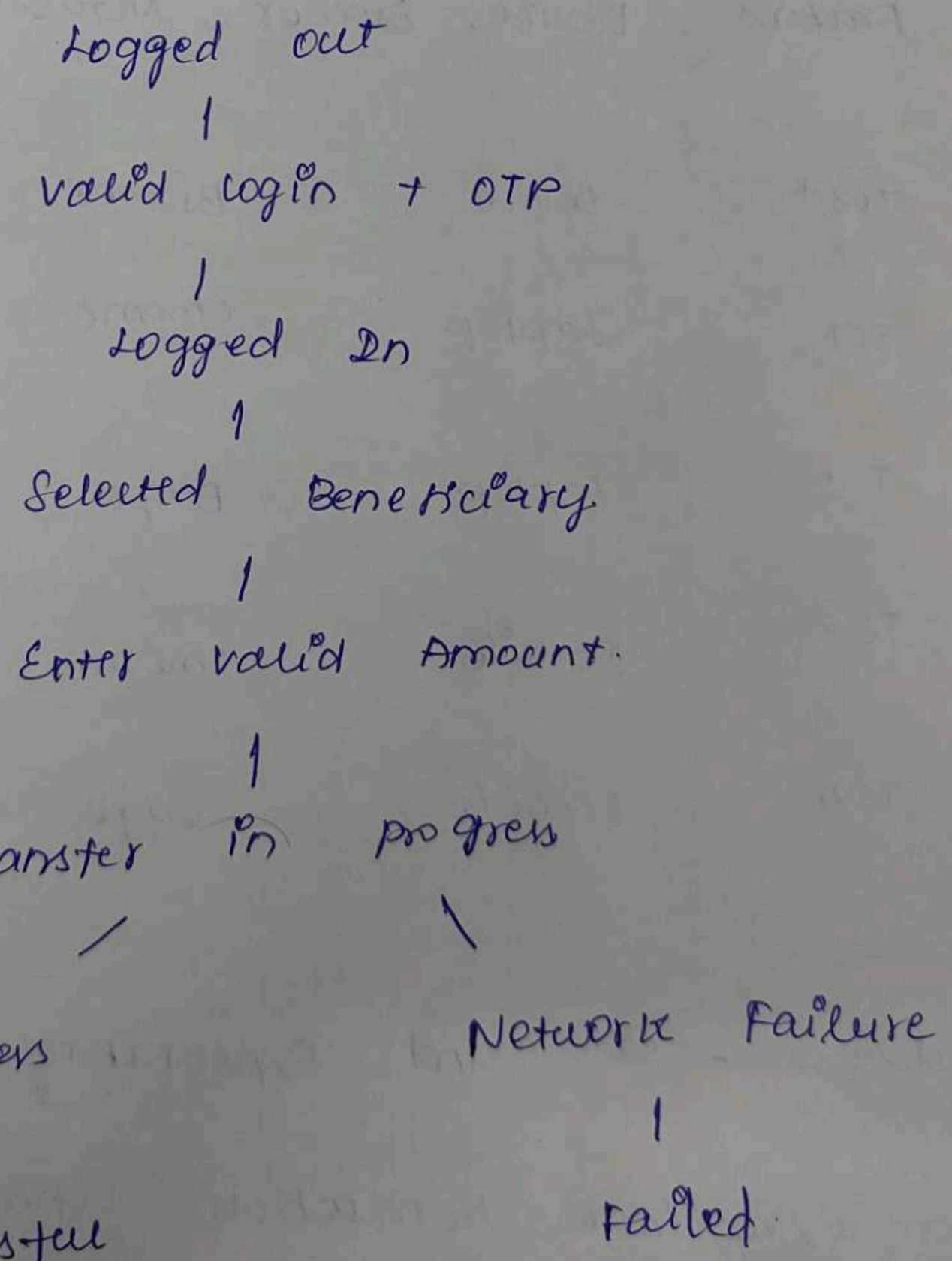
2) select valid, invalid, and boundary inputs.

3) check whether the expected output matches the actual output.

4) use statement and branch coverage to evaluate test completeness.

2) Online Banking application:-

a) State Transition Diagram:-



b) OTP Boundary Testing :-

Test case	OTP Attempts	Expected Result
TC 1	correct on first attempt	Login successfull
TC2	Two corong, then correct	Login successfull
TC3	Three consecutive wrong	Account locked.

c) Pairwise Testing :-

Factors : Device, Brower, Network.

Test	Device	Brower	Network
TC1	Laptop	chome	Wi-Fi
TC2	Laptop	Edge	Mobile Data
TC3	Mobile	chome	Mobile Data
TCH	Mobile	edge	Wi-Fi.

d) Use - case and Exploratory Testing :-

* Duplicate transaction : Check whether repeated clicks transfer money twice.

* session expiry : check whether an expired

session prevents transfers.

* Network failure : Interrupt the network during transfer.

* Wrong beneficiary : verify that money reaches the selected beneficiary

e) COTS payment Gateway Testing :

* Integration : verify payment requests and responses.

* Compatibility : Test across devices and browers.

* Reliability : chek failures, timeouts & retries.

Limitation :

* Internal code can't be inspected, so testing mainly relies on external behaviour and available documentation.

3) Smart ATM Fund - Transfer System :-

a) Control Flow Graph (CFG)

START

↓

Read Inputs

↓

Amount <= 0 ?

/ Yes

↓

Invalid Amount

\ NO

↓

Amount > Balance ?

/ Yes

↓

Insufficient

\ NO

↓

Amont > Daily Limit?

/ Yes

↓

Limit Error

\ NO

↓

Deduct Amount

↓

Transfer Succes

↓

STOP.

1. amout <= 0
2. amount > Balance
3. amout > daily limit.

b) cyclomatic complexity:

Formula:

$$V(G) = E - N + 2P.$$

E = edges, N = nodes, P = connected components.

There are 3 decision nodes:-

$$V(G) = 3+1 = 4.$$

c) Indepent paths and Test cases:-

Test	Amount	Balance	Daily limit	Expected output
TC1	-10	1000	500	Invalid Amt
TC2	1200	1000	1500	Insufficient Balance
TC3	600	1000	500	Daily limit Exceeded
TC4	200	1000	500	Tranfer Success.

d) Coverage Techniques:-

* Statement coverage: Execute every statement at least once

* Branch coverage: Test both True and False outcomes of each decision.

* Condition coverage: Test each condition as True or False

* Path coverage: Execute all independent paths

e) Test Adequacy :-

* 100% statement coverage doesn't guarantee defect-free software because some branches and conditions may remain untested.

4) Online login security system :-

a) Loop Decision structures :-

b) Loop Testing :-

Test	Password Input	Expected Result
TC1	correct first	Login Successful
TC2	wrong, correct second	Login Successful
TC3	wrong twice, correct third	Login Successful
TC4	Three wrong passwords	Account Locked
TC5	No password input	No Login attempt, verify input handling

c) Logical Detect and correction :-

```
count = 0
logged_in = false
while count < 3:
    password = input ("Enter password : ")
    if password == "admin":
        print ("login successful")
        logged_in = True
        break
    else:
        count += 1
if not logged_in:
    print ("Account locked")
```

Regression testing: Test correct password on attemp 1, 2 and 3, followed by Three incorrect password. only three incorrect attempts should produce the lock msg.

d) Coverage Techniques:

* Condition coverage: Test True and False conditions.

* Branch coverage: Test correct and incorrect passwords.

* path testing: Test saccess on attempts 1, 2 & 3 plus complete failure.

e) Mutation testing:

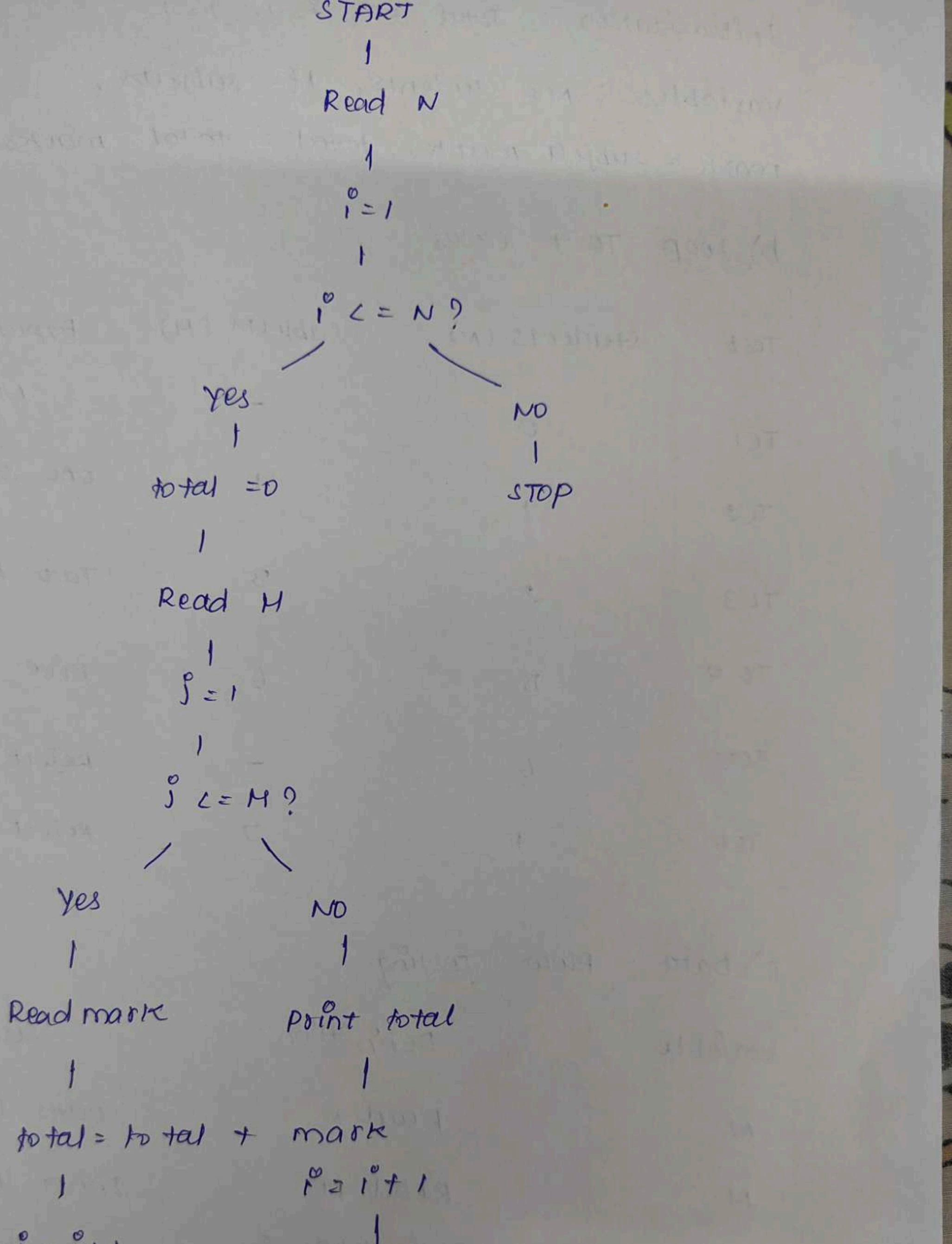
1. change count < 3 to count <4.

2. change count += 1 to count += 2

3. Remove the break statement.

5) Student Result Praising System:-

a) control Flow graph and variables:-



Outer loop : processes each student

Inner loop : Reads and adds each subject's marks

Initialization : Total = 0, i = 1, j = 1.

variables : N = students, M = subjects,

mark = subject mark, total = total marks.

b) Loop Test cases :-

Test	Students (N)	Subjects (M)	Expected Result
TC1	0	-	No results
TC2	1	1	One student is total displayed
TC3	2	3	Two total display
TC 4	5	6	Five totals display
TC5	6	-	Reject it max is 5
TC6	1	7	Reject it max is 6

c) Data - Flow Testing.

variable	Definition	Use
N	Read N	Outer loop condition
M	Read M	Inner loop condition
i	initialized / inc i	outer loop
j	Initialized / inc j	Inner loop
Mark	Read mark	Add to total
total	Initialized / update total	Print total

d) ECP, BVA, Error Guessing:

* ECP: Divide student counts, Subject counts, and marks into valid and invalid groups.

* BVA: Test values at the minimum and max limits.

* Error Guessing: Test negative marks, marks above the permitted max, zero subjects, and incorrect total

e) Test Adequacy:

* Statement coverage: Execute every statement.

* Branch coverage: Test zero, one, typical, and max iterations.

* Loop coverage: Test True, False loop condition.

* Data-Flow coverage: check variable Initialization, updates and uses.